BANKNOTE AUTHENTICATION USING LOGISTIC REGRESSION

(UCI BANKNOTE AUTHENTICATION DATASET)

A MACHINE LEARNING LAB MINI-PROJECT REPORT SUBMITTED TO

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in
Artificial Intelligence and Machine Learning

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CERTIFICATE

This is to certify that the mini-project work entitled **BANKNOTE AUTHENTICATION USING LOGISTIC REGRESSION** is a bonafide work carried out by **Vinayaka J (4NI24CI122)** in partial fulfillment for the award of degree of **Bachelor of Engineering in Artificial Intelligence and Machine Learning**, of Visvesvaraya Technological University, Belagavi, during the year **2025-26**. It is certified that all corrections / suggestions indicated during internal assessment have been incorporated and the corrected copy has been deposited in the department library. This mini-project report has been approved in partial fulfillment for the award of the said degree as per academic regulations of The National Institute of Engineering (Autonomous Institution).

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ABSTRACT

This mini-project builds and evaluates a **Logistic Regression** classifier for banknote authentication using the UCI Banknote Authentication dataset, which contains 1,372 records described by four wavelet-derived image features: Variance, Skewness, Curtosis and Entropy. The model predicts the binary `Class` label and is assessed on records that were not seen during training.

The data was explored for quality, class balance and feature ranges, split using an **80/20 stratified split** (`random_state=42`), and standardised with a `StandardScaler` fitted on the training data only to avoid data leakage. On the 275-record test set, the model achieved an accuracy of **97.09%**, with precision 0.9385, recall 1.0000 and F1-score 0.9683 for Class 1. Re-thresholding the predicted probabilities at 0.30, 0.50 and 0.70 showed that the 0.70 threshold removed seven of the eight false positives, raising accuracy to 99.64%. This is an educational exercise; the model is not certified for real banking, legal, currency-authentication or security decisions.

Contents

Acknowledgment

Abstract

1 PROBLEM STATEMENT AND DATASET **1**

1.1 Problem Statement 1

1.2 Dataset Description 1

1.3 Label Convention 1

2 EXPLORATORY DATA ANALYSIS **2**

2.1 Shape and Data Quality 2

2.2 Class Distribution and Feature Ranges 2

2.3 Visualisations 3

2.4 Observations 4

3 METHODOLOGY **5**

3.1 Feature and Target Preparation 5

3.2 Train-Test Split 5

3.3 Feature Scaling 5

3.4 Model Training and Evaluation 5

4 RESULTS **6**

4.1 Prediction Behaviour 6

4.2 Performance Metrics 6

4.3 Confusion Matrix 7

4.4 Threshold Comparison 8

4.5 Optional Extensions 8

5 ANSWERS TO REPORT QUESTIONS **9**

Conclusion **11**

Chapter 1

PROBLEM STATEMENT AND DATASET

1.1 Problem Statement

A bank wants to support the screening of banknotes using measurements extracted from their images. This project builds and evaluates a Logistic Regression classifier that uses four numerical features (Variance, Skewness, Curtosis, Entropy) to predict the binary class label in the dataset, and assesses how well it classifies records not seen during training. This is an educational exercise; the model is not certified for real banking, legal, currency-authentication or security decisions.

1.2 Dataset Description

The UCI Banknote Authentication dataset (V. Lohweg, 2012; DOI 10.24432/C55P57) contains 1,372 records, four continuous input features computed from 400×400 grey-scale banknote images with a wavelet transform, and one binary target, `Class`. There are no missing values.

1.3 Label Convention

The label convention is not stated in the official documentation. The official UCI page says the images were taken from genuine and forged banknote-like specimens, but its variables table gives no description of Class 0 and Class 1. Secondary sources disagree (some state 1 = genuine, others 0 = genuine). This report therefore does not depend on the mapping for any computed result.

For discussion only, the working assumption is **Class 0 = genuine, Class 1 = forged**, with Class 1 as the positive class. This should be confirmed with faculty; if reversed, the roles of false positives and false negatives swap (see Q27 and Q29 in Chapter 6).

Chapter 2

EXPLORATORY DATA ANALYSIS

2.1 Shape and Data Quality

The dataset has 1,372 rows × 5 columns; all four features are `float64` and `Class` is `int64`; there are 0 missing values. There are 24 duplicate rows (35 rows involved), all in Class 0. Duplicates were kept: the dataset is small, repeated specimens are plausible, and removing them would alter the official dataset; their effect is noted as a limitation.

2.2 Class Distribution and Feature Ranges

Class 0 contains 762 records (55.54%) and Class 1 contains 610 records (44.46%), so the dataset is mildly imbalanced, but not severely. Ranges differ across features: Skewness spans about −13.8 to 13.0 (std 5.87) whereas Entropy spans about −8.5 to 2.5 (std 2.10), which motivates scaling. Class means differ clearly in Variance (2.28 vs −1.87) and Skewness (4.26 vs −0.99).

Table 2.1: Descriptive statistics of the four input features

Statistic	Variance	Skewness	Curtosis	Entropy
count	1372.000	1372.000	1372.000	1372.000
mean	0.434	1.922	1.398	−1.192
std	2.843	5.869	4.310	2.101
min	−7.042	−13.773	−5.286	−8.548
25%	−1.773	−1.708	−1.575	−2.413
50%	0.496	2.320	0.617	−0.587
75%	2.821	6.815	3.179	0.395
max	6.825	12.952	17.927	2.450

2.3 Visualisations

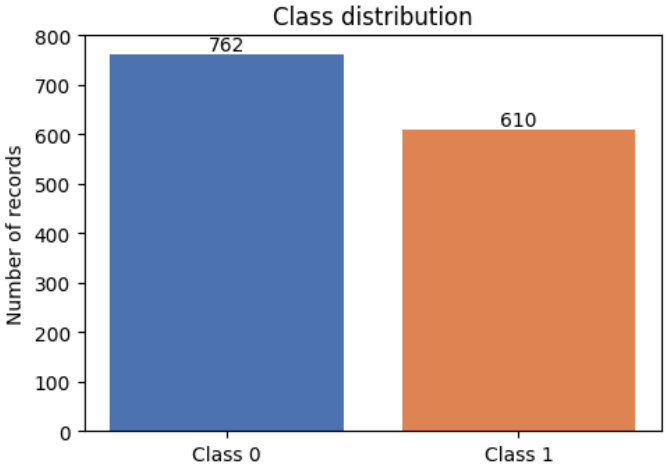


Figure 2.1: Class distribution (Class 0 = 762, Class 1 = 610)

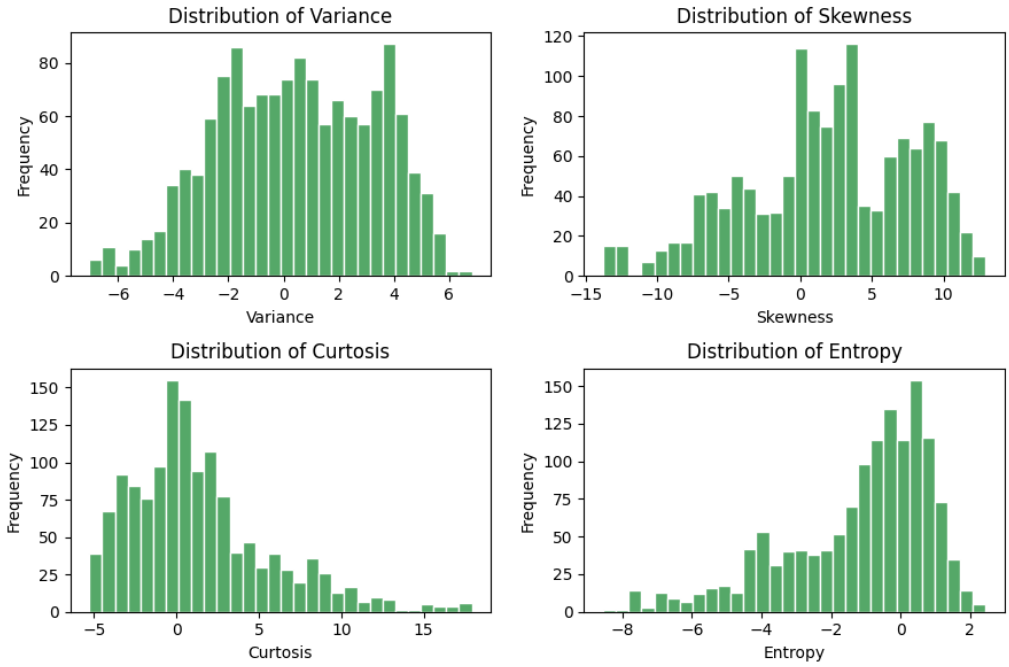


Figure 2.2: Histograms of the four input features

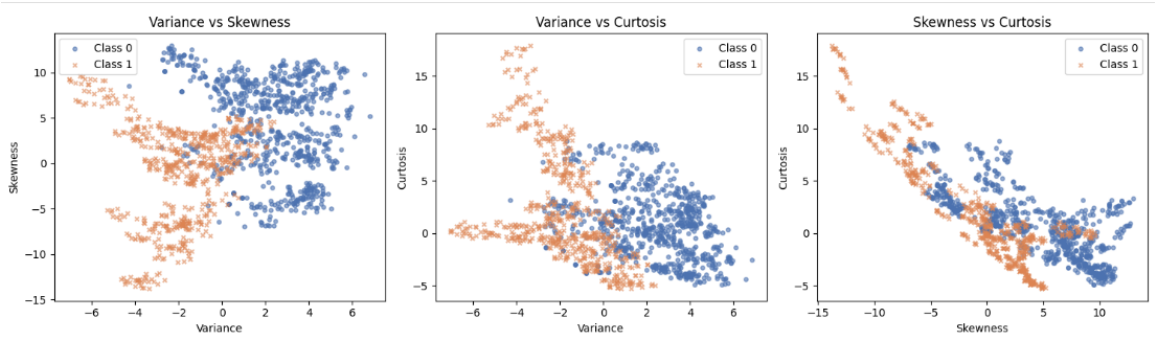


Figure 2.3: Class-coloured scatter plots (Class 0 = blue circles, Class 1 = orange crosses)

2.4 Observations

1. The class bar chart confirms the 762 vs 610 imbalance.
2. In Variance vs Skewness the classes form largely separate clouds (Class 0 at higher Variance, Class 1 at lower) but overlap near the boundary, so no single feature guarantees correct classification.
3. Variance vs Curtosis and especially Skewness vs Curtosis overlap much more, so Curtosis alone separates the classes poorly.
4. Histograms show Curtosis with a long upper tail and Entropy skewed to the left, on very different scales.

Chapter 3

METHODOLOGY

3.1 Feature and Target Preparation

The input matrix is X = (Variance, Skewness, Curtosis, Entropy) with shape 1372×4, and the target is y = `Class` with shape 1372.

3.2 Train-Test Split

An **80/20 stratified split with** `random_state=42` gave 1,097 training records (609 Class 0, 488 Class 1) and 275 testing records (153 Class 0, 122 Class 1). The Class-1 share is 44.48% in training and 44.36% in testing.

3.3 Feature Scaling

`StandardScaler` was **fitted on the training data only** and then applied to both sets, to avoid data leakage.

3.4 Model Training and Evaluation

`LogisticRegression(max_iter=1000)` (default $C = 1.0$, lbfgs solver) was trained on the scaled training data and converged in 11 iterations. Labels and probabilities (`predict_proba`) were produced for the test set, evaluated at the default 0.50 threshold, and re-thresholded at 0.30, 0.50 and 0.70.

The scaled coefficients are given below; Entropy contributes little.

Table 3.1: Scaled model coefficients

Term	Coefficient
Variance	−4.67
Skewness	−4.81
Curtosis	−4.40
Entropy	0.26
Intercept	−1.40

Chapter 4

RESULTS

4.1 Prediction Behaviour

The results below are on the test set of 275 records, with Class 1 as the positive class. The notebook lists actual vs predicted labels for 15 records and probabilities for 10. All of the first 15 are correct, and most probabilities are close to 0 or 1. Overall there are **8 errors out of 275**, all Class 0 predicted as Class 1, with $P(\text{Class } 1)$ between 0.55 and 0.73 — the model is least sure on these borderline records.

4.2 Performance Metrics

Table 4.1: Test-set metrics at threshold 0.50 (Class 1 positive)

Metric (Class 1 positive)	Value
Accuracy	0.9709 (97.09%)
Precision	0.9385
Recall	1.0000
F1-score	0.9683

Table 4.2: Classification report

Class	Precision	Recall	F1	Support
0	1.0000	0.9477	0.9732	153
1	0.9385	1.0000	0.9683	122
accuracy			0.9709	275
macro avg	0.9692	0.9739	0.9707	275
weighted avg	0.9727	0.9709	0.9710	275

4.3 Confusion Matrix

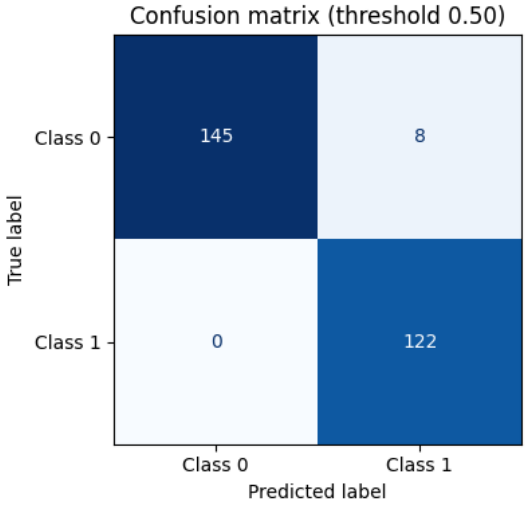


Figure 4.1: Confusion matrix at threshold 0.50

At threshold 0.50, TN = 145, FP = 8, FN = 0 and TP = 122. The layout is [[TN, FP], [FN, TP]] with labels ordered [0, 1], verified from scikit-learn. Of 130 notes predicted Class 1, 122 are truly Class 1 (precision 0.9385); all 122 actual Class 1 notes were found (recall 1.0000).

Under the working assumption (Class 1 = forged), a **false negative** (forged note accepted as genuine) would be the more concerning error, and this model made none on the test set; its 8 false positives are genuine notes wrongly flagged.

4.4 Threshold Comparison

Table 4.3: Effect of the decision threshold on test-set performance

Thr.	Pred. 1	Acc.	Prec.	Recall	F1	TN	FP	FN	TP
0.30	130	0.9709	0.9385	1.0000	0.9683	145	8	0	122
0.50	130	0.9709	0.9385	1.0000	0.9683	145	8	0	122
0.70	123	0.9964	0.9919	1.0000	0.9959	152	1	0	122

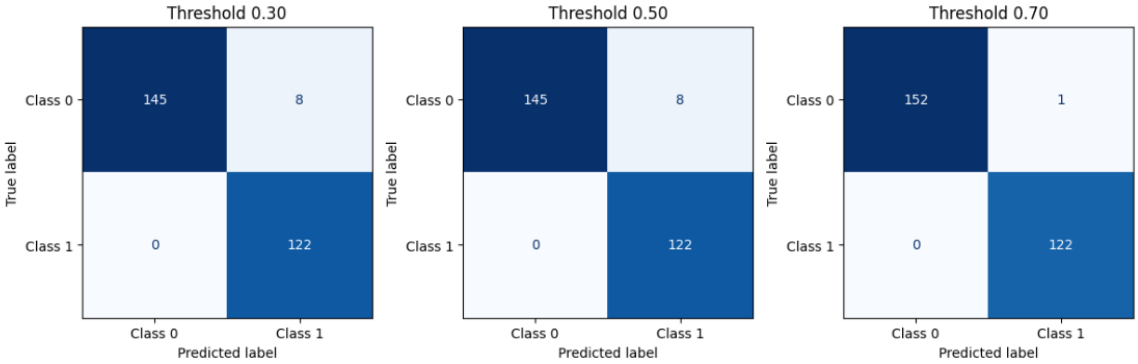


Figure 4.2: Confusion matrices at thresholds 0.30, 0.50 and 0.70

Interpretation. Thresholds 0.30 and 0.50 give identical results because no test record has $P(\text{Class } 1)$ between 0.30 and 0.50. Raising the threshold to 0.70 reduces Class 1 predictions from 130 to 123 and removes 7 of the 8 false positives, lifting precision to 0.9919 and accuracy to 99.64%, while recall stays 1.0000 on this test set. In general a higher threshold trades recall for precision and a lower one does the opposite; the choice should reflect the cost of each error, not just the highest accuracy. With only 275 test records the apparent advantage of 0.70 should not be over-interpreted, and on other data a high threshold could miss borderline Class 1 notes.

4.5 Optional Extensions

The following additional experiments were run in the notebook:

- 5-fold stratified cross-validation accuracy = 0.9811 ± 0.0063.
- Decision Tree test accuracy 0.9927 vs Logistic Regression 0.9709.
- Smaller C (stronger regularisation) lowers accuracy, e.g. 0.8582 at $C = 0.001$ vs 0.9855 at $C \geq 10$.

Chapter 5

ANSWERS TO REPORT QUESTIONS

Q14. Regression vs classification. Regression predicts a continuous number (e.g. a price); classification predicts a discrete category (e.g. genuine or forged).

Q15. Why "regression" in the name? It fits a linear combination of the features, $z = b_0 + b_1 x_1 + \ldots$, and passes it through the sigmoid to obtain a probability, which is then thresholded into a class. The linear-model heritage gives the name; the output use is classification.

Q16. Input features. Variance, Skewness, Curtosis (spelt as in the dataset) and Entropy.

Q17. Target and labels. `Class`, a binary integer (0 or 1). The official UCI documentation says the specimens are genuine and forged but does not define which value is which. Working assumption used here: 0 = genuine, 1 = forged (unverified).

Q18. Size, missing, duplicates. 1,372 records, 4 features plus the target; 0 missing values; 24 duplicate rows (35 involved), all Class 0, kept in the data.

Q19. Balanced? Mildly imbalanced: 762 (55.54%) vs 610 (44.46%). Not severe, but accuracy alone is not enough.

Q20. Why split train/test? To estimate performance on unseen data; testing on training data rewards memorisation and overstates real performance.

Q21. Stratified splitting. It keeps class proportions about equal in both subsets (Class-1 share 44.48% train vs 44.36% test), so the evaluation is not skewed by an unlucky split.

Q22. Why scale? Features have different ranges (e.g. Skewness std 5.87 vs Entropy 2.10). Scaling helps the optimiser converge and makes regularisation and coefficient comparison fair.

Q23. Data leakage. Leakage is when information from test data influences training. Fitting the scaler before the split lets the test records affect its mean and standard deviation, so the test set is no longer truly unseen and metrics may look better than reality.

Q24. Sigmoid. $\sigma(z) = 1/(1+e^{-z})$ maps any real score to a value between 0 and 1, interpreted as $P(\text{Class } 1)$.

Q25. Probability 0.80. The model estimates an 80% chance of Class 1 given the features; it is not a guarantee, and about 1 in 5 such cases could be Class 0 if the model is well calibrated.

Q26. Metrics in own words. *Accuracy:* share of all predictions that are correct. *Precision:* of records flagged positive, how many truly are. *Recall:* of all true positives, how many were found. *F1:* harmonic mean of precision and recall; low if either is low.

Q27. TP/TN/FP/FN here (positive = Class 1, assumed forged). TP = 122: forged correctly flagged. TN = 145: genuine correctly accepted. FP = 8: genuine wrongly flagged. FN = 0: forged wrongly accepted. If Class 1 is actually genuine, reverse the real-world meanings of FP and FN.

Q28. Effect of threshold. $0.30 \rightarrow 0.50$: no change (accuracy 0.9709, precision 0.9385, recall 1.0000, F1 0.9683; FP 8, FN 0). $0.50 \rightarrow 0.70$: accuracy 0.9964, precision 0.9919, F1 0.9959, recall unchanged at 1.0000; FP falls from 8 to 1.

Q29. Metric to prioritise. If failing to identify a forged note is especially costly, prioritise recall for the forged class (minimise false negatives), while monitoring precision/F1 to keep false alarms manageable. Under the working assumption forged = Class 1, so that is recall for Class 1 (1.0000 here). If forged were actually Class 0, the corresponding measure would be recall for Class 0 (0.9477 at threshold 0.50, 0.9935 at 0.70).

Q30. Limitations.

(a) Small dataset with 24 duplicates and a single split, giving a noisy estimate;
(b) features come from controlled specimen images, so performance may not carry to real-world scans, other currencies or new counterfeiting methods;
(c) the linear model may miss non-linear structure (a Decision Tree scored higher on this split: 99.27%);
(d) label meaning is not documented officially;
(e) probability calibration was not checked.

CONCLUSION

Logistic Regression trained on four standardised wavelet features classified the 275 held-out banknote records with 97.09% accuracy. For Class 1, the positive class, precision was 0.9385, recall 1.0000 and F1 0.9683, with 145 true negatives, 8 false positives, no false negatives and 122 true positives. Because accuracy hides which errors occur, precision, recall and the confusion matrix are more informative. In a screening context where letting a forged note pass is the costly error, recall for the forged class is the most important metric; this assumes Class 1 is forged, a mapping the official UCI page does not state and which must be confirmed.

Thresholds 0.30 and 0.50 gave identical results because no test record had a probability between them, while 0.70 removed seven of eight false positives, raising precision to 0.9919 and accuracy to 99.64% without losing recall on this test set. That gain should not be over-interpreted, since higher thresholds can reduce recall on other data.

Limitations include a small dataset with 24 duplicate rows, a single train-test split, pre-extracted features from controlled specimens and a linear model. The model is educational and unsuitable for real banking use.

References

[1] Lohweg, V. (2012). *Banknote Authentication* [Dataset]. UCI Machine Learning Repository. https://doi.org/10.24432/C55P57

[2] *scikit-learn documentation*. https://scikit-learn.org